Independent review: weighted C0 branch

1 November 2026

Fresh isolated Lean replay passed eight modules: fourteen main declarations, seven producer controls and six independent controls. All 27 axiom readbacks use only propext, Classical.choice and Quot.sound. The scoped field formulas agree with the inspected native source.

Formula correspondence

Native `projective_law_gate.rs`, normalize lines 533–566, decodes degree-one coordinates as x/z^2 and $y/(z^5w)$. The weighted addition branch, lines 899–1023, crosses the other input's weights. With $Z = z_1z_2$ and $W = w_1w_2$, the checked scaling is

$$B_c = Z^8B, \quad A_c = Z^9WA, \quad C_c = Z^7WC.$$

The literal output obeys

$$x_o = Z^{18}W^2x_n, \quad y_o = Z^{45}W^5y_n, \quad z_o = Z^9WB, \quad w_o = 1.$$

Decoding cancels the weights. Reconstruction covers arbitrary stored inputs with nonzero weights, and the B/C guard equivalences retain their explicit nonzero-factor assumptions.

Conditional curve membership

Composition requires a field, four nonzero input weights, $B \neq 0$, $C = 0$, a monic degree-five polynomial with coefficient four zero, and both input Mumford residual divisibilities. It concludes nonzero output weights and $y^2 = f(x)$. Divisor-class addition, native caller validity, machine words, counters, scheduling and compiler correctness remain separate bridges.

Independent boundary controls

The six independent controls cover mixed-sign nonunit weights with decoded $(10, 291)$; output $z = 20 \cdot 21^9$; missing-w normalization failure; zero-w singular output; arbitrary raw reconstruction; and nonzero C. They use kernel tactics rather than native evaluation. A reviewer initially expected factor 10 instead of 20 in the output-z control. That failed attempt is preserved; only the reviewer's expectation changed. No producer repair occurred.

Evidence and process

`run.sh`, module receipts, `before.sha256`, `after.txt` and `reviewed-source.sha256` bind the local replay. Retained dependency binaries remain a trust boundary; there was no complete dependency rebuild or independent kernel binary check.

The reviewer read native source and producer status/plan before writing a local source-first plan, then inspected proof bodies. This was not fully blind or precommitted. Review dispatch preceded producer publication; T-1225 records that sequencing deviation and publication binding. Workflow completion is not asserted.

`REVIEW.md` and `status.json` provide source references, assumptions, failures and scope. No scientific run or new measurement was performed; no cost ratio is inferred.